But we know that $a \notin \overline{A}$.

**Theorem:** Let $(X, \tau)$ be a topological space, A and B be arbitrary subsets of X. Then we have the following:

(i) A is closed $\Leftrightarrow \overline{A} = A$

(ii) $\overline{\overline{A}} = \overline{A}$

(iii) If $A \subseteq B$ then $\overline{A} \subseteq \overline{B}$

(iv) $\overline{A}$ is the smallest closed subset of X containing A.

(v) $\mathring{A} = X \setminus \overline{X \setminus A}$

(vi) $\text{ext } A = X \setminus \overline{A}$

(vii) $\partial A = \overline{A} \cap \overline{X \setminus A}$

(viii) $\dot{A} \subseteq \mathring{\overline{A}}$

(ix) $\text{ext } A = \text{ext}(\overline{A})$

(x) $\partial \overline{A} \subseteq \partial A$

(xi) $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$

(xii) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

**Proof:**

(i) $A$ is closed $\iff X \setminus A \in \tau$

$\iff ext(A) = X \setminus A$

$\iff \overline{A} = X \setminus ext(A)$

$= X \setminus (X \setminus A)$

$= A$

(ii) Recall that $\overline{A}$ is a closed set.

Thus $\overline{A} = \overline{\overline{A}}$ by (i)

(iii) $A \subseteq B \Rightarrow X \setminus B \subseteq X \setminus A$

$\Rightarrow (X \setminus B)^\circ \subseteq (X \setminus A)^\circ$

$\Rightarrow ext\, B \subseteq ext\, A$

$\Rightarrow X \setminus ext\, A \subseteq X \setminus ext\, B$

$\Rightarrow \overline{A} \subseteq \overline{B}$.

(iv) Let $K$ be a closed subset of $X$ containing $A$, i.e. $A \subseteq K$.

$\Rightarrow A \subseteq \overline{A} \subseteq \overline{K} = K$ by (iii) & (i)

(v) $X \setminus \overline{X \setminus A}$

$= X \setminus (X \setminus ext(X \setminus A))$

$= ext(X \setminus A)$

$= (X \setminus (X \setminus A))^\circ = \mathring{A}$

(vi) Since $\overline{A} = X \setminus ext(A)$

$$ext(A) = X \setminus (X \setminus ext(A)) = X \setminus \overline{A}.$$

(vii) $\overline{A} \cap \overline{X \setminus A}$

$$= (X \setminus ext(A)) \cap (X \setminus ext(X \setminus A))$$
$$= (X \setminus ext(A)) \cap (X \setminus (X \setminus (X \setminus A))^\circ)$$
$$= (X \setminus ext(A)) \cap (X \setminus \mathring{A})$$
$$= X \setminus (\mathring{A} \cup ext(A)) \text{ by de Morgan's Law}$$
$$= \partial A.$$

(ix) $ext(\overline{A}) = (X \setminus \overline{A})^\circ$

$$= X \setminus \overline{A}$$
$$= ext(A) \quad \text{by (vi)}.$$

(x) $\partial \overline{A} = \overline{\overline{A}} \cap \overline{X \setminus \overline{A}}$ by (vii)

$$= \overline{A} \cap \overline{X \setminus \overline{A}}$$
$$\subseteq \overline{A} \cap \overline{X \setminus A} = \partial A \quad \text{by (vii)}$$

(xi) $A \cap B \subseteq A, B \Rightarrow \overline{A \cap B} \subseteq \overline{A}, \overline{B}$

$$\Rightarrow \overline{A \cap B} \subseteq \overline{A} \cap \overline{B}.$$

(xii) $A, B \subseteq A \cup B$

$\Rightarrow \overline{A}, \overline{B} \subseteq \overline{A \cup B}$

$\Rightarrow \overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$ (1)

Since $A \subseteq \overline{A}$ & $B \subseteq \overline{B}$,

$A \cup B \subseteq \overline{A} \cup \overline{B}$

$\Rightarrow \overline{A \cup B} \subseteq \overline{\overline{A} \cup \overline{B}} = \overline{A} \cup \overline{B}$ (2)

By (1) & (2) $\overline{A} \cup \overline{B} = \overline{A \cup B}$.

Note: For arbitrary sets A & B

$\overline{A} \cap \overline{B}$ need not be contained in $\overline{A \cap B}$:

Let $(\mathbb{R}, \tau)$ be the usual topological space, $A = (1, 2)$ & $B = (2, 5)$.

Then $\overline{A \cap B} = \overline{\phi} = \phi$. However

$\overline{A} \cap \overline{B} = [1, 2] \cap [2, 5] = \{2\}$.

$\overline{A} \cap \overline{B} \not\subseteq \overline{A \cap B}$.

Example: Let $X = \{a, b, c, d\}$, $\tau = \{\phi, \{a, b, c\}, \{d\}, X\}$ & $A = \{a, b\} \subseteq X$. Determine the following subsets of $X$: $\mathring{A}$, $ext(A)$, $\partial A$, $\overline{A}$, $(\overline{A})^{\circ}$, $ext(\overline{A})$ & $\partial \overline{A}$.

Soln

$\mathring{A} = \phi$.

$X \setminus A = \{c, d\} \Rightarrow ext(A) = (X \setminus A)^{\circ}$

$= \{d\}$

$\partial A = X \setminus (\mathring{A} \cup ext(A))$

$= X \setminus \{d\}$

$= \{a, b, c\}$

$\overline{A} = X \setminus ext(A) = \{a, b, c\}$

$(\overline{A})^{\circ} = \{a, b, c\} = \overline{A}$

$ext(\overline{A}) = ext(A) = \{d\}$

$\partial \overline{A} = X \setminus ((\overline{A})^{\circ} \cup ext(\overline{A}))$

$$= X \setminus \{a, b, c\} \cup \{d\}$$

$$= X \setminus X = \phi.$$

We infer from this example that, in general $\partial \overline{A} \neq \partial A$ & that $\overline{\overset{\circ}{A}} \neq \overset{\circ}{A}$.

Here we can also find $\overline{A}$ as follows:

Find $\tau_K$ = the class of $\tau$-closed subsets of $X$:

$\tau_K = \{X, \{d\}, \{a, b, c\}, \phi\}$

Now $\overline{A}$ is the smallest closed subset of $X$ that contains $A$.

So we have $\overline{A} = \{a, b, c\}$.

Definition: Let $(X, \tau)$ be a topological space & $A \subseteq X$. If $\overline{A} = X$ then $A$ is said to be dense in $X$.

**Proposition:** Let $(X, \tau)$ be a topological space, & $\mathcal{B}$ a base for $\tau$ & $A \subseteq X$. Then $A$ is a dense subset of $X$, i.e. $\overline{A} = X \iff A \cap B \neq \emptyset$ for every $B \in \mathcal{B}$.

**Proof:** Exercise.

**Examples:**

1. Let $(X, \tau)$ be an indiscrete topological space. Find all possible dense subsets of $X$.

**Sol$^n$** Let $A \subseteq X$. Because $X \neq \emptyset$

$\tau = \{\emptyset, X\}$, we have only one nonempty open set, namely $X$ itself. But then $A \cap X = A \neq \emptyset$

$\Rightarrow \overline{A} = X$ by the previous proposition.

$\therefore$ Any nonempty subset of $X$ is a dense subset of $X$.

2. Let $(X, \tau)$ be a discrete topological space. Find all possible dense subsets of $X$.

Soln

If $A$ is a subset of $X$ then we know that $A$ is a closed (& also open) set. Therefore $\overline{A} = A$. Hence $A$ is a dense subset of $X \iff A = X$.

3. Let $(\mathbb{R}, \tau)$ be the usual topological space. Then the set $\mathbb{R} \setminus \mathbb{Q}$ of all irrational numbers is a dense subset of $\mathbb{R}$:

Recall that $\mathcal{B} = \{(a,b) \mid a, b \in \mathbb{R}, a < b\}$ is a base for $\tau$. Moreover any open interval $(a,b) \in \mathcal{B}$ contains at least one irrational number. Thus $\mathbb{R} \setminus \mathbb{Q} \cap (a,b) \neq \emptyset$ for any $(a,b) \in \mathcal{B}$. By the previous

proposition $\overline{\mathbb{R} \setminus \mathbb{Q}} = \mathbb{R}$.